SAFETY DATA SHEET

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Version 1

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identification

Product Code/Catalogue Number:	984391
SDS Number:	D14434_SDS_Cholesterol Food Standard _EN
Product Name	**Cholesterol Food standard**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.

1.3. Details of the supplier of the safety data sheet

Company	**Thermo Fisher Scientific Oy** Analyzers & Automation Clinical Diagnostics Ratastie 2, P.O. Box 100 FI-01621 Vantaa, Finland
Telephone number	+358 10 329200
E-mail address	system.support.fi@thermofisher.com

1.4. Emergency telephone number

CHEMTREC INTERNATIONAL +1 703-741-5970

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Skin Corrosion/irritation - Category 2
Serious Eye Damage/Eye Irritation - Category 2
Skin Sensitization - Category 1
Classification according to EU Directives 67/548/EEC or 1999/45/EC
Xn - Harmful.
R22 - Harmful if swallowed.

2.2. Label elements

Signal Word Warning

Hazard Statements
H315 - Causes skin irritation
H319 - Causes serious eye irritation
H317 - May cause an allergic skin reaction
Precautionary Statements
P280 - Wear protective gloves/ protective clothing/ eye protection/ face protection
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and

easy to do. Continue rinsing

2.3. Other hazards
No information available

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Component	Weight %	CLP Classification - Regulation (EC) No 1272/2008	DSD Classification - 67/548/EEC
Cholan-24-oic acid, 3,7,12-trihydroxy-, (3.alpha.,5.beta.,7.alpha.,12.alpha.)- (CAS #: 81-25-4)	1 - 10 %	Acute Tox. 4 (H312) Skin Irrit. 2 (H315) Skin Sens. 1 (H317) Eye Irrit. 2 (H319)	Xn; R22 Xi; R36/38
Poly(oxy-1,2-ethanediyl), .alpha.-isotridecyl-.omega.-hydroxy- (CAS #: 9043-30-5)	1 - 3 %	Acute Tox 4 (H302) Eye Dam. 1 (H318)	Xn; R22 R41
Tris (hydroxymethyl) aminomethane (CAS #: 77-86-1)	1 - <2 %	Skin Irrit. 2 (H315) Eye Irrit. 2 (H319) STOT SE 3 (H335)	Xi; R36/37/38
2-Chloroacetamide (CAS #: 79-07-2)	0.1 - <1 %	Acute Tox. 3 (H301) Skin Sens. 1 (H317) Repr. 2 (H361f)	T; R25 R43 Repr.Cat.3; R62
Sodium azide (CAS #: 26628-22-8)	<0.1 %	Acute Tox. 2 (H300) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410) (EUH032)	T+; R28 R32 N; R50-53

For the full text of the R-phrases and H-Statements mentioned in this Section, see Section 16.

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

Inhalation
Move to fresh air. Oxygen or artificial respiration if needed.

Skin Contact
Wash off immediately with plenty of water. If skin irritation persists, call a physician.

Eye Contact
Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.

Ingestion
Clean mouth with water. Call a physician immediately.

4.2. Most important symptoms and effects, both acute and delayed
No information available.

4.3. Indication of any immediate medical attention and special treatment needed
Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water. Carbon dioxide (CO2). Foam.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

Hazardous Combustion Products

None under normal use conditions.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Avoid contact with the skin and the eyes.

6.2. Environmental precautions

Prevent further leakage or spillage if safe to do so. Prevent product from entering drains.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Avoid contact with skin and eyes. Ensure adequate ventilation.

7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep at temperatures between 2° and 8 °C.

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Component Exposure Limits

Component	Finland	European Union	The United Kingdom	Germany
2-Chloroacetamide				Haut
Sodium azide	TWA: 0.1 mg/m³ 8 tunteina STEL: 0.3 mg/m³ 15 minuutteina Iho	Skin TWA 0.1 mg/m³ STEL 0.3 mg/m³	Skin TWA 0.1 mg/m³ STEL 0.3 mg/m³	MAK 0.2 mg/m³ (inhalable)

Component	Sweden	Norway	Denmark	France
Sodium azide	STV: 0.3 mg/m³ 15 minuter LLV: 0.1 mg/m³ 8 timmar. Hud	Hud Ceiling: 0.3 mg/m³	TWA: 0.1 mg/m³ 8 timer Hud	TWA / VME: 0.1 mg/m³ (8 heures). restrictive limit STEL / VLCT: 0.3 mg/m³. restrictive limit Peau

8.2. Exposure controls

Engineering Measures

Ensure adequate ventilation, especially in confined areas. Avoid contact with skin, eyes and clothing.

Personal protective equipment

Eye Protection Safety glasses with side-shields (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Disposable gloves	See manufacturers recommendations	-	EN 374	(minimum requirement)

Inspect gloves before use.
Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)
Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.
Remove gloves with care avoiding skin contamination.

Skin and body protection
Long sleeved clothing

Respiratory Protection No protective equipment is needed under normal use conditions.

Small scale/Laboratory use
Maintain adequate ventilation No personal respiratory protective equipment normally required
Recommended half mask:-

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls
No information available.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Appearance	Colorless	
Physical State	Liquid	
Odor	Characteristic	
Odor Threshold	No data available	
pH	8.5 @ 25°C	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No data available	
Flash Point	No data available	**Method -** No information available
Evaporation Rate	No data available	
Flammability (solid,gas)	No information available	
Explosion Limits	No data available	
Vapor Pressure	No data available	
Vapor Density	No data available	(Air = 1.0)
Specific Gravity / Density	No data available	
Bulk Density	No data available	
Water Solubility	No information available	
Solubility in other solvents	No information available	

Partition Coefficient (n-octanol/water)

Component	log Pow
Cholan-24-oic acid, 3,7,12-trihydroxy-, (3.alpha.,5.beta.,7.alpha.,12.alpha.)-	3.38

Autoignition Temperature	No data available
Decomposition Temperature	No data available
Viscosity	No data available
Explosive Properties	No information available
Oxidizing Properties	No information available

9.2. Other information

No data available

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity
No data available

10.2. Chemical stability
Stable under normal conditions

10.3. Possibility of hazardous reactions
No information available.
No information available.

10.4. Conditions to avoid
Heat.

10.5. Incompatible materials
Metals.

10.6. Hazardous decomposition products
None under normal use conditions.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on toxicological effects

Product Information
Product does not present an acute toxicity hazard based on known or supplied information
(a) acute toxicity;

Oral	Based on available data, the classification criteria are not met
Dermal	Based on available data, the classification criteria are not met
Inhalation	Based on available data, the classification criteria are not met

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Cholan-24-oic acid, 3,7,12-trihydroxy-, (3.alpha.,5.beta.,7.alpha.,12.alpha.)-	4950 mg/kg (Rat)		
Poly(oxy-1,2-ethanediyl), .alpha.-isotridecyl-.omega.-hydroxy-	1000 mg/kg (Rat)		
Tris (hydroxymethyl) aminomethane	5900 mg/kg (Rat)		
2-Chloroacetamide	138 mg/kg (Rat)		
Sodium azide	27 mg/kg (Rat)	50 mg/kg (Rat) 20 mg/kg (Rabbit)	

(b) skin corrosion/irritation;
Category 2.
(c) serious eye damage/irritation;
Category 2.

(d) respiratory or skin sensitization;
Respiratory
Not Classified.
Skin
Category 1.

(e) germ cell mutagenicity;
Not Classified

(f) carcinogenicity;
Not Classified

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;
Based on available data, the classification criteria are not met.

(h) STOT-single exposure;
Based on available data, the classification criteria are not met.

(i) STOT-repeated exposure;
Not Classified.

Target Organs
No information available.

(j) aspiration hazard;
Not Classified.

Symptoms / effects,both acute and delayed
No information available

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Component	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Cholan-24-oic acid, 3,7,12-trihydroxy-, (3.alpha.,5.beta.,7.alpha.,12.alpha.)-	LC50 = 70,29 mg/l 96h	LC50 = 75.52 mg/L 48h		
2-Chloroacetamide				EC50 = 10.3 mg/L 5 min EC50 = 19.5 mg/L 15 min EC50 = 31.7 mg/L 30 min
Sodium azide	5.46 mg/L LC50 96 h 0.7 mg/L LC50 96 h 0.8 mg/L LC50 96 h			

12.2. Persistence and degradability
No information available

12.3. Bioaccumulative potential
No information available

Component	log Pow	Bioconcentration factor (BCF)
Cholan-24-oic acid, 3,7,12-trihydroxy-, (3.alpha.,5.beta.,7.alpha.,12.alpha.)-	3.38	No data available

12.4. Mobility in soil
No information available

12.5. Results of PBT and vPvB assessment
No data available for assessment.

12.6. Other adverse effects
None known

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues / Unused Products
Dispose of in accordance with local regulations.

Contaminated Packaging
Dispose of in accordance with local regulations.

SECTION 14: TRANSPORT INFORMATION

	IMDG/IMO	ADR	IATA
	Not regulated	Not regulated	Not regulated
14.1. UN number	-	-	-
14.2. UN proper shipping name	-	-	-
14.3. Transport hazard class(es)	-	-	-
14.4. Packing group	-	-	-

14.5. Environmental hazards
No hazards identified

14.6. Special precautions for user
No special precautions required

14.7. Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code
Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

Component	EINECS	ELINCS	NLP	TSCA	DSL	NDSL	PICCS	ENCS	IECSC	AICS	KECL
Cholan-24-oic acid, 3,7,12-trihydroxy-, (3.alpha.,5.beta.,7.alpha.,12.alpha.)-	201-337-8	-		X	X	-	-	X	X	X	-
Poly(oxy-1,2-ethanediyl), .alpha.-isotridecyl-.omega.-hydroxy-	-	-	>1<2.5 mol ethoxylated units	X	X	-	X	X	X	X	X
Tris (hydroxymethyl) aminomethane	201-064-4	-		X	X	-	X	X	X	X	X
2-Chloroacetamide	201-174-2	-		X	X	-	X	X	X	X	X
Sodium azide	247-852-1	-		X	X	-	X	X	X	X	X

National Regulations

Component	Germany - Water Classification (VwVwS)	Germany - TA-Luft Class
Cholan-24-oic acid, 3,7,12-trihydroxy-, (3.alpha.,5.beta.,7.alpha.,12.alpha.)-	WGK 2	
Tris (hydroxymethyl) aminomethane	WGK 2	
2-Chloroacetamide	WGK 2	
Sodium azide	WGK 2	

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H300 - Fatal if swallowed
H301 - Toxic if swallowed
H302 - Harmful if swallowed
H312 - Harmful in contact with skin
H315 - Causes skin irritation
H317 - May cause an allergic skin reaction
H318 - Causes serious eye damage
H319 - Causes serious eye irritation
H335 - May cause respiratory irritation
H361f - Suspected of damaging fertility
H400 - Very toxic to aquatic life
H410 - Very toxic to aquatic life with long lasting effects
EUH032 - Contact with acids liberates very toxic gas

Full text of R-phrases referred to under sections 2 and 3

R22 - Harmful if swallowed
R25 - Toxic if swallowed
R28 - Very toxic if swallowed
R32 - Contact with acids liberates very toxic gas
R41 - Risk of serious damage to eyes
R43 - May cause sensitization by skin contact
R50 - Very toxic to aquatic organisms
R53 - May cause long-term adverse effects in the aquatic environment
R62 - Possible risk of impaired fertility
R36/37/38 - Irritating to eyes, respiratory system and skin
R36/38 - Irritating to eyes and skin

Legend

CAS - Chemical Abstracts Service
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
PNEC - Predicted No Effect Concentration
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - Volatile Organic Compounds

Key literature references and sources for data

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

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Reason for revision	Update to CLP Format.

Disclaimer

The information provided on this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its

publication. The information given is designed only as a guide for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered as a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other material or in any process, unless specified in the text.